

EDUCATION

Massachusetts Institute of Technology

May 2025

Bachelor of Science in Artificial Intelligence & Decision Making

Cambridge, MA

- Relevant Classes: *Algorithms and Data Structures, Graduate-level Machine Learning, TinyML & Efficient Deep Learning Computing, Linear Algebra, Modelling with Machine Learning, Probability & Random Variables*

EXPERIENCE

Meta, Zero Trust Networking

July 2025 –

Software Engineer

New York City, NY

- Built and shipped a CLI tool to evaluate forward-proxy egress policies locally (service/user identity → allow/deny + rule explanation), eliminating the need for slow production canary tests.
- Built an **AI-driven agent** for HTTP-onboarding that automates forward-proxy client setup (VIP provisioning + DNS configuration), reducing end-to-end onboarding time from ~2+ weeks to a few hours.
- Leading the migration of forward-proxy destination filters to a modern policy evaluation engine (APE), decoupling policy logic from the proxy runtime and making it reusable as a standalone library for other clients/services to adopt.

Meta, Infra + Ranking: Ads Serving Platform

May 2024 – August 2024

Software Engineering Intern

New York City, NY

- Improved DB API service reliability: Fixed broken/flaky test processes for high-traffic databases(up to 37.3M reads daily)
- Deprecated stale cost collector objects, freeing up wasted compute and memory and decluttering the service.
- Cost collector onboarding: **Wrote data ingestion and augmentation pipelines** to process teams' and platforms' granular resource use metrics to generic MW cost metrics and into a UI dashboard.
- Contributed documentation to the FBGold wiki and mentored a new-hire with ramp-up tasks.

Broad Institute of Harvard & MIT, Cimini Lab

September 2023 – May 2024

Eric and Wendy Schmidt Center Funded Research and Innovation Scholar

Cambridge, MA

- Researched and implemented **optimization methods** to bring inference engines for **biological image segmentation models to the browser** with ONNX.
- **Customizing and fine-tuning the U-Net & SAM model** and their variants.

Meta, Pytorch Distributed Team

May – August 2023

Software Engineering Intern

Menlo Park, CA

- Enabled training using the **pipeline parallel distributed paradigm** for more robust training in a multi-GPU environment by implementing functionality to save model checkpoints.
- **Refactored legacy modules within the PyTorch Distributed repository**, improving code maintainability and reducing technical debt.

PROJECTS

Samaki.ai

March 2026

- Engineered an automated aquaculture monitoring system utilizing a **custom YOLOv12 object detection pipeline** to track fish health metrics, growth rates, and behavioral patterns in real-time.
- **Designed and integrated custom feeder hardware** with the computer vision inference engine, enabling closed-loop, automated feeding cycles driven by live visual telemetry.

ArchiGen

March – May 2023

- Designed and **built a stable-diffusion based generative AI text-to-floor-plans platform** as part of an on-campus AI projects incubator.
- **Preprocessed the RPLAN dataset(80k samples)** and fine-tuned a general stable-diffusion model on floor plans.

TECHNICAL SKILLS

Languages & Frameworks: Python, C++, C, Rust, Go, PyTorch, ONNX, Django, React, Node.js

Tools & Infrastructure: Claude code, RAG, LoRA fine-tuning, Git, RESTful APIs, Distributed Systems, Hyperparameter Tuning, CAD, Tmux, Nvim